School Worksheet

Class 12 Mathematics, Chapter 1: Relations and Functions

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

Let R be the relation on the set $\{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. Then R is:

(A) reflexive, but neither symmetric nor transitive

(B) an equivalence relation

(C) symmetric and transitive only

(D) transitive only

2 MCQ — 1 mark

The number of equivalence relations on the set $\{1, 2, 3\}$ containing $(1, 2)$ and $(2, 1)$ is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

3 MCQ — 1 mark

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is:

(A) one-one and onto **(B)** many-one and onto

(C) one-one but not onto **(D)** neither one-one nor onto

4 MCQ — 1 mark

The function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(x) = 2x$ is:

(A) bijective

(B) injective but not surjective

(C) surjective but not injective

(D) neither injective nor surjective

5 MCQ — 1 mark

If set A has 3 elements and set B has 4 elements, then the number of one-one functions from A to B is:

(A) 12 **(B)** 24 **(C)** 64 **(D)** 81

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** The relation $R = \{(a, b) : a \leq b\}$ on the set of real numbers is not an equivalence relation. **Reason (R):** The relation $R = \{(a, b) : a \leq b\}$ is not symmetric. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3 - 4x$ is a bijection.	2 marks

Level 2

19 marks

8	Short answer Show that the relation R on the set of integers $\mathbb{Z}$ defined by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation.	2 marks
9	Short answer Check whether the relation R on $\mathbb{R}$ defined by $R = \{(a, b) : a \leq b^3\}$ is reflexive, symmetric or transitive.	3 marks
10	Short answer Let $A = \{1, 2, 3, \ldots, 9\}$ and R be the relation on $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $a + d = b + c$. Prove that R is an equivalence relation and find the equivalence class $[(2, 5)]$.	3 marks
11	Short answer Let $f : \mathbb{N} \to \mathbb{N}$ be defined by $f(n) = \begin{cases} \frac{n+1}{2}, & n \text{ odd} \\ \frac{n}{2}, & n \text{ even} \end{cases}$. Show that f is onto but not one-one.	3 marks
12	Short answer Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$ is neither one-one nor onto.	3 marks
13	Short answer Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Show that $f : A \to B$ defined by $f(x) = \frac{x - 2}{x - 3}$ is one-one and onto.	3 marks
14	Short answer Let $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(a, b) : b = a + 1\}$ on A. Is R reflexive, symmetric or transitive? Justify.	2 marks

Q No.	Question	Marks

Level 3 — 25 marks

15 Short answer — 3 marks

Let L be the set of all lines in a plane and $R = \{(L_1, L_2) : L_1 \perp L_2\}$. Show that R is symmetric but neither reflexive nor transitive.

16 Long answer — 5 marks

Let $\mathbb{N}$ be the set of natural numbers and R be the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\, R\, (c, d)$ if and only if $ad = bc$. Show that R is an equivalence relation.

17 Long answer — 5 marks

Show that the function $f : \mathbb{R} \to \{x \in \mathbb{R} : -1 < x < 1\}$ defined by $f(x) = \dfrac{x}{1 + |x|}$ is one-one and onto.

18 Long answer — 5 marks

Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$. Show that $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$ is an equivalence relation on A. Find the set of all elements related to 1 and write the equivalence class $[2]$.

19 Short answer — 3 marks

Show that the function $f : [-1, 1] \to \mathbb{R}$ defined by $f(x) = \dfrac{x}{x + 2}$ is one-one. Also find the range of f.

20 Case study — 4 marks

Case Study: A classroom teacher wants to assess her students' understanding of "relations". She writes the following five relations, each defined on the set $A = \{1, 2, 3\}$:
$R_1 = \{(2, 3), (3, 2)\}$, $R_2 = \{(1, 2), (1, 3), (3, 2)\}$, $R_3 = \{(1, 2), (2, 1), (1, 1)\}$,
$R_4 = \{(1, 1), (1, 2), (3, 3), (2, 2)\}$, $R_5 = \{(1, 1), (1, 2), (3, 3), (2, 2), (2, 1), (2, 3), (3, 2)\}$.

(i) Identify the relation which is reflexive, transitive but not symmetric. — 1 mark

(ii) Identify the relation which is reflexive and symmetric but not transitive. — 1 mark

(iii)(a) Identify the relations which are symmetric but neither reflexive nor transitive. — 2 marks

OR

(iii)(b) What pairs should be added to the relation R_2 to make it an equivalence relation? — 2 marks

Answer key

Chapter 1: Relations and Functions, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) reflexive, but neither symmetric nor transitive
2	(B) 2
3	(D) neither one-one nor onto
4	(B) injective but not surjective
5	(B) 24
6	(A) Both A and R are true, and R is the correct explanation of A
7	f is one-one and onto, so f is a bijection.

Level 2

8	R is reflexive, symmetric and transitive, hence an equivalence relation.
9	R is neither reflexive, nor symmetric, nor transitive.
10	R is an equivalence relation; $[(2,5)] = \{(1,4),(2,5),(3,6),(4,7),(5,8),(6,9)\}$.
11	f is onto but not one-one.
12	f is neither one-one nor onto.
13	f is one-one and onto.
14	R is neither reflexive, nor symmetric, nor transitive.

Level 3

15	R is symmetric but neither reflexive nor transitive.
16	R is reflexive, symmetric and transitive, hence an equivalence relation.
17	f is one-one and onto.
18	R is an equivalence relation; elements related to 1: $\{1,5,9\}$; $[2] = \{2,6,10\}$.
19	f is one-one; range of $f = \left[-1, \frac{1}{3}\right]$.
20	(i) R_4 (ii) R_5 (iii)(a) R_1 and R_3 OR (iii)(b) add $(1,1),(2,2),(3,3),(2,1),(3,1),(2,3)$